Project Design Phase
Problem – Solution Fit Template

Date	18 February 2026
Team ID	LTVIP2026TMIDS65644
Project Name	Rising Waters: A Machine Learning Approach To Flood Prediction Output: A Next-Generation Approach to Flood Prediction
Maximum Marks	2 Marks

Problem – Solution Fit Template: Wind Turbine Energy

- *Rising Waters* is a machine learning-driven project that predicts flood risk levels by analyzing rainfall patterns, meteorological data, and environmental features. The web-based system enables authorities and communities to receive timely flood risk indications, thereby facilitating early intervention and mitigation. Users input location-specific environmental and weather data through a **Flask web interface**, and the system classifies flood risk into three categories: **Low, Mid, or High**.

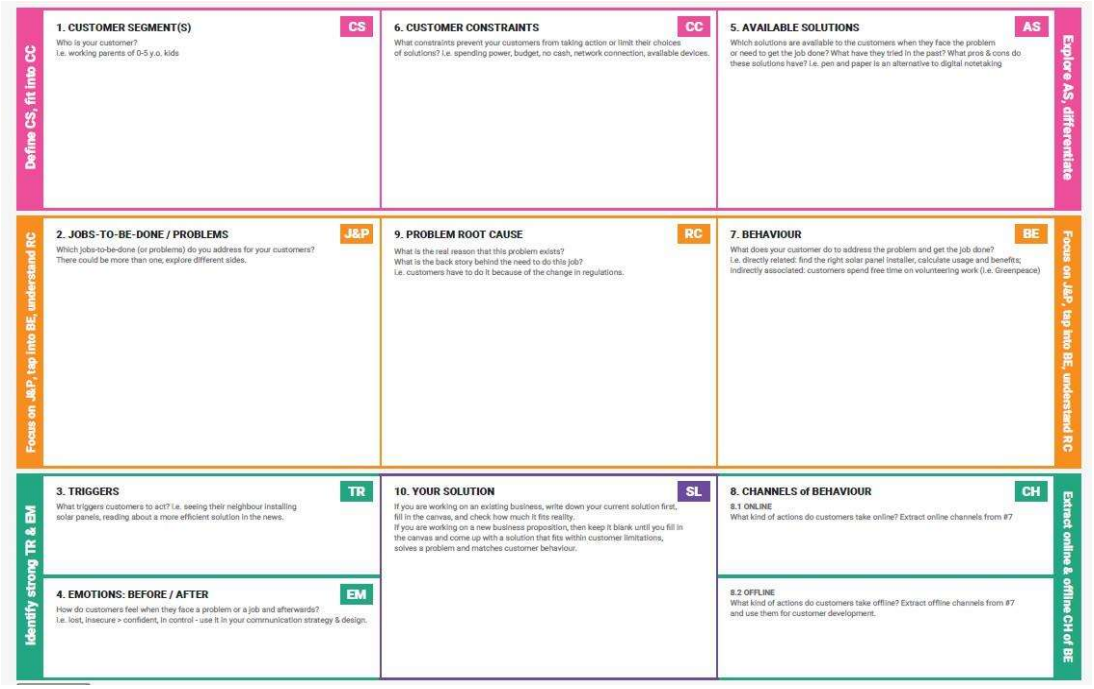
Solution

"Rising Waters" is a machine learning project designed to predict flood likelihood by analyzing rainfall, environmental, and geographical data. The aim is to provide early warnings and support decision-making to reduce flood-related hazards. The system combines predictive modeling with a web-based interface to deliver real-time flood risk assessments.

Purpose:

The primary purpose of "Rising Waters" is to use machine learning for early flood prediction by analyzing historical and real-time environmental data to provide timely warnings, reduce risks to life and property, and improve disaster management decisions

Template:



References:

1. https://www.ideahackers.network/problem-solution-fit-canvas/
2. https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe